

Rearranging the terms and removing repeated terms:

$$(A+B+C+D) \cdot (A+B+C+D)' \cdot (A+B+C+D) \cdot (A+B+C+D)' \cdot (A+B+C+D) \cdot (A+B+C+D)' \cdot (A+B+C+D) \cdot (A+B+C+D)'$$

$$\Pi(0,1,2,3,5,9,13,15)$$

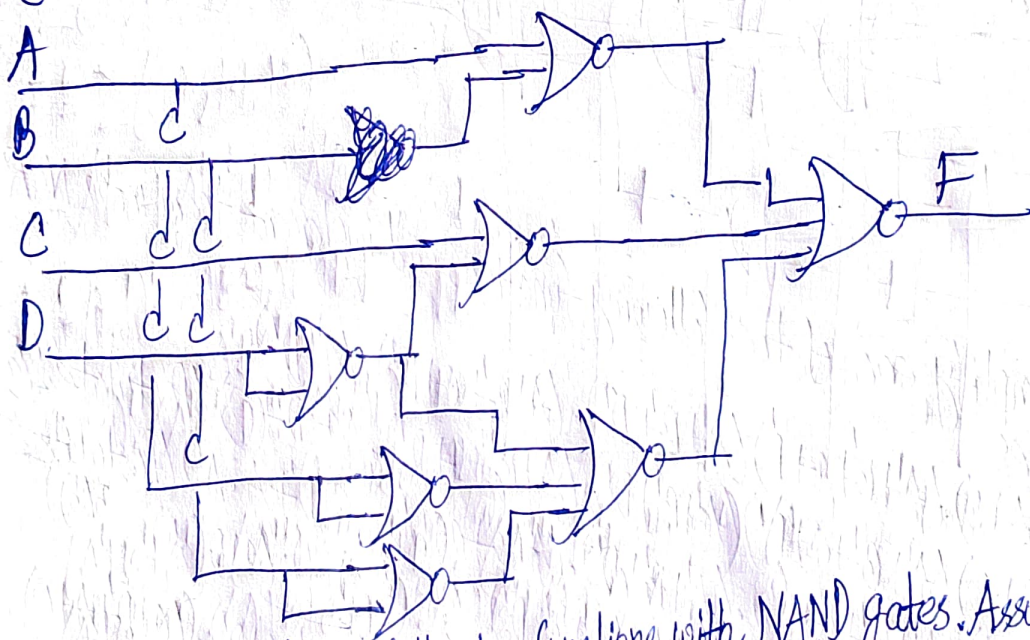
AB \ CD	00	01	11	10
00	1	0	0	0
01	0	0	0	0
11	0	1	1	0
10	0	0	0	0

$$F: (A+B) \cdot (C+D)' \cdot (A+B'D)'$$

$$((A+B) \cdot (C+D)' \cdot (A+B'D)')'$$

$$= ((A+B)' + (C+D)' + (A+B'D)')' \quad (\text{Theorem 3})$$

(Extended Thm 5(b))



3-11: Implement the following functions with NAND gates. Assume that both the normal and complement inputs are available.

b) $(AB+A'B')(CD'+C'D)$ with two input-gates.

$$= ((AB+A'B')')' ((CD'+C'D)')' \quad (\text{Theorem 3})$$

$$= ((AB)')' ((A'B)')' ((CD)')' ((C'D)')' \quad (\text{Theorem 5(a)})$$

$$= ((AB)')' ((A'B)')' ((CD)')' ((C'D)')' \quad (\text{Theorem 3}) \quad \square$$

a) $BD + BCD + AB'C'D + A'B'CD'$ with no more than six gates each having 3 inputs

$$= BD + BCD + AB'C'D + A'B'CD' \quad (\text{Postulate 2(b)})$$

$$= BD + BCD + D'(AB'C + A'B'C) \quad (\text{Postulate 4(a)})$$

$$= BD + D'(AB'C + A'B'C) \quad (\text{Thm 6(a)})$$